

Que. What are the three main layers of software architecture? Write one line about each.

Ans.

- **Presentation Layer (UI Layer):** Handles the user interface and displays information to users while accepting their input.
- **Business Logic Layer (Application Layer):** Processes data, applies business rules, and manages the application's core functionality.
- **Data Layer (Database Layer):** Stores, retrieves, and manages data from the database for the application.

Que. Define client and server in your own words with one real-life example.

Ans.

Client: A client is a device or application that sends a request to access a service or data.

Server: A server is a computer or system that receives the client's request, processes it, and sends back the required data or service.

Real-life example: When you open YouTube on your phone, your phone is the **client**, and YouTube's computers are the **servers** that send the videos to your phone.

Que. Write 2 differences each between desktop apps, web apps, and mobile apps.

Ans. **Desktop Apps**

1. Installed directly on a computer or laptop.
2. Can work without an internet connection (for many applications).

Web Apps

1. Run in a web browser without installation.
2. Usually require an internet connection to use.

Mobile Apps

1. Installed on smartphones or tablets.
2. Designed for touch screens and mobile operating systems like Android or iOS.

Que. Name any 3 databases and mention whether each is SQL or NoSQL.

Ans.

- MySQL – SQL database.
- PostgreSQL – SQL database.
- MongoDB – NoSQL database.